ITIL V4 Foundation

Module 4: ITIL Management Practices

Comprehensive Student Notes

Complete Guide covering 50+ Pages

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In this module:

- ✓ All 34 ITIL Practices Overview
- ✓ Detailed Analysis of 15 Core Practices
- ✓ Incident vs. Problem vs. Service Request
- ✓ Change Enablement & Deployment Management
- ✓ Service Desk Models & Best Practices
- ✓ Extensive Real-world Case Studies

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Introduction & Overview

Module Introduction

Welcome to Module 4, the final and most practical section of your ITIL 4 Foundation journey. While previous modules established the theoretical framework (Service Value System, Four Dimensions), this module focuses on the actual **practices**—the sets of organizational resources designed for performing work or accomplishing objectives.

In simple terms: Module 3 taught you the "why" and the "strategy"; Module 4 teaches you the "what" and the "how." These practices are the tools you will use every day in your IT career. From the moment a user calls the Service Desk to the complex resolution of a major problem, practices guide your actions.

> **EVOLUTION FROM PROCESSES TO PRACTICES**
>
> In ITIL v3, these were called "Processes" (e.g., Incident Management Process). In ITIL 4, they are called "Practices." This is a significant shift. A **Process** is just a set of steps. A **Practice** is much broader—it includes the process, but also the people, skills, tools, and partners required to execute it effectively. It is a holistic capability.

This comprehensive module covers the 34 ITIL management practices, with a deep dive into the 15

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most critical practices that appear most frequently on the exam and in daily work. We will use detailed case studies to show how these practices prevent disasters and create value.

Why Practices Matter

- **Consistency:** Practices ensure that similar issues are handled in similar ways, reducing errors.
- **Efficiency:** Optimized practices reduce waste and time (e.g., automating password resets).
- **Reliability:** Practices like Availability Management ensure systems stay online.
- **Value Creation:** Practices support the Service Value Chain activities (Plan, Improve, Engage, Design & Transition, Obtain/Build, Deliver & Support).

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The 34 ITIL Practices Overview (M4.L1-L2)

ITIL 4 identifies 34 management practices, categorized into three groups based on their origin and focus.

Figure 4.1: The Three Categories of ITIL Practices

1. General Management Practices (14)

These practices have been adopted and adapted for service management from general business management domains.

▪ Architecture Management ▪ Continual Improvement ▪ Information Security Management ▪ Knowledge Management ▪ Measurement and Reporting ▪ Organizational Change Management ▪ Portfolio Management	▪ Project Management ▪ Relationship Management ▪ Risk Management ▪ Service Financial Management ▪ Strategy Management ▪ Supplier Management ▪ Workforce and Talent Management

2. Service Management Practices (17)

These practices have been developed in service management and ITSM industries.

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▪ Availability Management ▪ Business Analysis ▪ Capacity and Performance Management ▪ Change Enablement ▪ Incident Management ▪ IT Asset Management ▪ Monitoring and Event Management ▪ Problem Management ▪ Release Management	▪ Service Catalog Management ▪ Service Configuration Management ▪ Service Continuity Management ▪ Service Design ▪ Service Desk ▪ Service Level Management ▪ Service Request Management ▪ Service Validation and Testing

3. Technical Management Practices (3)

These practices focus on technology and have been adapted from technology management domains.

- Deployment Management
- Infrastructure and Platform Management
- Software Development and Management

EXAM TIP: CATEGORIZATION

You may be asked to identify which category a specific practice belongs to.
Hint: If it sounds like general business (e.g., Project Management, Risk Management), it's General. If it's specific to IT services (e.g., Incident, Service Desk), it's Service. If it's purely technical (e.g., Software Development), it's Technical.

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Detailed Practices Coverage

Practice 1: Service Desk (M4.L3)

Definition and Purpose

> **DEFINITION: SERVICE DESK**
>
> The practice of capturing demand for incident resolution and service requests. It should also be the entry point and single point of contact for the service provider with all of its users.

The Service Desk is not just a "Help Desk." It is the strategic interface between the IT organization and the users. It creates the first impression of IT service quality.

Key Activities

- **Incident Capture:** Logging reports of things that are broken (e.g., "My email is down").
- **Service Request Capture:** Logging requests for new things (e.g., "I need a new laptop").
- **Information Provision:** Answering questions (e.g., "How do I change my password?").
- **Communication:** Broadcasting alerts about outages or maintenance.

Service Desk Models

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Organizations can structure their Service Desk in different ways depending on their needs.

Model	Description	Pros/Cons
Local Service Desk	Located physically close to the users it supports (e.g., in the same office building).	**Pros:** Personal touch, understands local context/culture. **Cons:** Expensive, duplication of resources.
Centralized Service Desk	A single location serving all users globally or regionally.	**Pros:** Cost-effective, consistent processes. **Cons:** May lack local cultural understanding/language skills.
Virtual Service Desk	Agents work from anywhere (remote), connected by technology. Allows "Follow-the-Sun."	**Pros:** 24/7 coverage, work-from-home flexibility. **Cons:** Requires robust technology and management.

DETAILED CASE STUDY: EUROPEAN BANK SERVICE DESK CRISIS

Scenario: A major European bank experienced a technical glitch slowing down online banking. Within minutes, thousands of calls flooded the Service Desk.

The Response:

- The Service Desk did not just log tickets. They became the **Crisis Communication Hub**.
- They activated an IVR (interactive voice response) message: "We are aware of the slowness and are fixing it." This deflected 40% of calls.
- They reassured anxious customers that their money was safe (Empathy/Customer

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Experience).
- They fed real-time impact data to the technical teams ("It's affecting Germany most severely").

Outcome: The technical team fixed the issue in 2 hours. Customer trust was maintained because the Service Desk provided clear, calm communication. The CEO credited the Service Desk with saving the bank's reputation.

Modern Trends

- **Omnichannel Support:** Users can contact IT via phone, email, chat, mobile app, or walk-up.
- **AI and Chatbots:** Handling routine queries (password resets) to free up humans for complex issues.
- **Self-Service:** Portals where users can solve their own problems.

KEY TAKEAWAY

The Service Desk is about **Customer Experience (CX)**. A technical fix is useless if the user felt ignored or disrespected during the process.

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Practice 2: Incident Management (M4.L4)

Definition and Purpose

DEFINITION: INCIDENT MANAGEMENT

The practice of minimizing the negative impact of incidents by restoring normal service operation as quickly as possible.

CRITICAL DISTINCTION: INCIDENT VS. PROBLEM

Incident: Dealing with the symptom to restore service NOW.
Problem: Dealing with the root cause to prevent recurrence LATER.
Do not confuse these. Incident Management is about speed, not long-term fixes.

Key Activities

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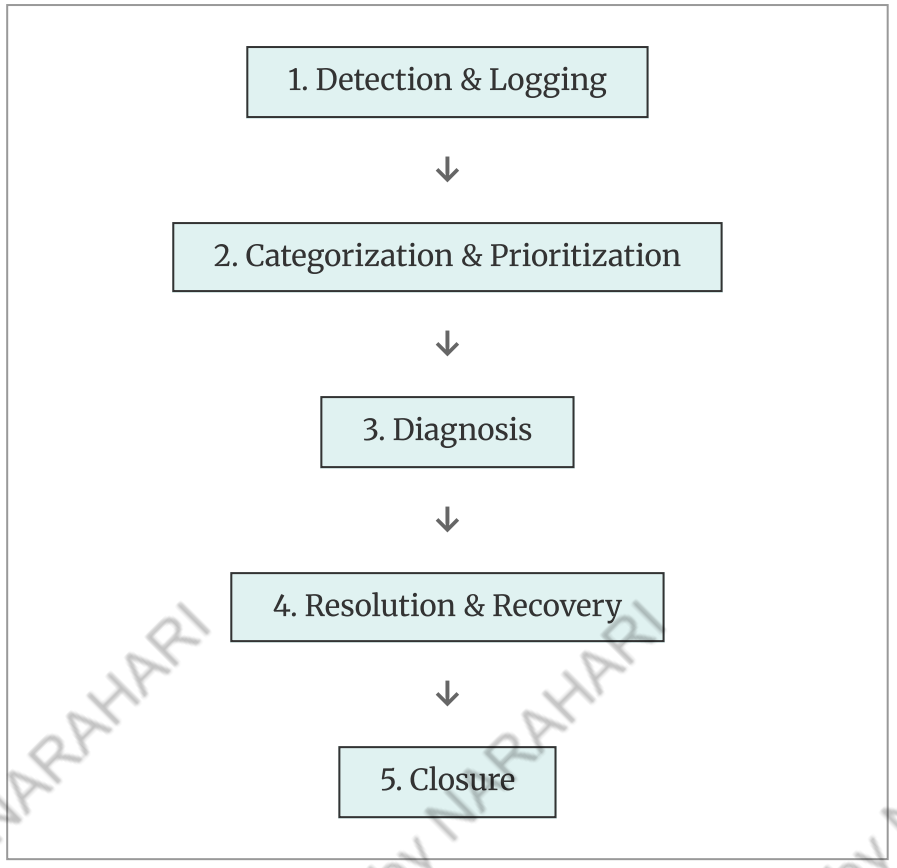
Figure 4.2: Incident Management Workflow



- **Logging:** Every incident must be logged. No "off the books" fixes.
- **Prioritization:** Based on **Impact** (how many users?) and **Urgency** (how quickly does business lose money?).
- **Escalation:**
  - *Functional Escalation:* Sending to a team with more skills (Level 1 to Level 2).
  - *Hierarchical Escalation:* Informing management (e.g., for Major Incidents).

DETAILED CASE STUDY: 2024 CROWDSTRIKE GLOBAL INCIDENT

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The Event: A faulty software update caused 8.5 million Windows computers to crash (Blue Screen of Death). Airlines, hospitals, and banks stopped functioning.

Incident Management in Action:

- **Mature Organizations:** Activated Major Incident Protocols immediately. They didn't try to debug the code; they focused on *restoration*. They identified workarounds (booting in Safe Mode, deleting the file) and deployed teams to apply them manually or via scripts. Recovery time: Hours.
- **Immature Organizations:** Panicked. Queued calls to Microsoft/CrowdStrike without internal triage. No prioritization of critical vs. non-critical servers. Recovery time: Days or Weeks.

Lesson: The goal was restoration. Finding *why* the code failed was a Problem Management task for later. Incident Management's job was simply to get the planes flying again.

Swarming

A modern approach where, instead of passing a ticket up a chain (Tier 1 -> Tier 2 -> Tier 3), a "swarm" of experts gathers immediately to look at the issue together. This is faster for complex issues.

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Practice 3: Problem Management (M4.L5)

Definition and Purpose

DEFINITION: PROBLEM MANAGEMENT

The practice of reducing the likelihood and impact of incidents by identifying actual and potential causes of incidents, and managing workarounds and known errors.

Problem Management Phases

1. **Problem Identification:** Analyzing trends (e.g., "Why does the server crash every Tuesday?").
2. **Problem Control:** Investigating the root cause. Documenting "Known Errors."
3. **Error Control:** Finding a permanent fix (often requires Change Control).

Figure 4.3: Incident vs. Problem Relationship

Incident Management	Problem Management
"Fix it fast!" Apply workaround. Restore service. *Example: Reboot server.*	"Fix it forever!" Find root cause. Apply permanent fix. *Example: Patch memory leak so reboot isn't needed.*

DETAILED CASE STUDY: TELECOM PASSWORD RESET PROBLEM

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Scenario: A telecom company's Service Desk was overwhelmed. 40% of all incidents were "Forgot Password."

Incident Management Response: Efficiently reset the passwords manually. (Fast, but repetitive).

Problem Management Investigation: Why so many resets?

- **Root Cause Analysis:** Policy required 15-character complex passwords changing every 30 days. Users couldn't remember them.
- **Solution:**
  - Implement Self-Service Password Reset tool (Tech fix).
  - Change policy to 90-day rotation (Policy fix).

Result: Incident volume dropped by 65%. $600K annual savings.

Workarounds and Known Errors

- **Known Error:** A problem that has been analyzed but not yet resolved.
- **Workaround:** A temporary way to overcome difficulties (e.g., "Reboot every night"). Problem Management creates workarounds for Incident Management to use.

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Practice 4: Service Request Management (M4.L6)

Definition and Purpose

> **DEFINITION: SERVICE REQUEST MANAGEMENT**
>
> The practice of supporting the agreed quality of a service by handling all pre-defined, user-initiated service requests in an effective and user-friendly manner.

Service Request vs. Incident

This is a critical exam topic.

- **Incident:** Something is broken. (Unplanned). E.g., "My laptop won't turn on."
- **Service Request:** I want something new. (Planned/Standard). E.g., "I want a new monitor."

Key Principles

- **Standardization:** Requests should be pre-defined in a catalog.
- **Automation:** Whenever possible, automate fulfillment (e.g., software installs).
- **Authorization:** Requests often need approval (financial or security).

> **DETAILED CASE STUDY: RETAIL CORP TRANSFORMATION**

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Before: Users emailed IT for everything ("I need access to the sales folder"). IT manually checked with managers, then manually granted access. Average time: 4.3 days.

After (Service Request Management):

- Created a **Self-Service Portal** (Catalog).
- Users select "Access to Sales Folder."
- System automatically emails Manager for approval button click.
- Once approved, system automatically adds user to Active Directory group.

Result: Average time: 1.7 hours. IT staff freed from manual work.

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Practice 5: Service Level Management (M4.L7)

Definition and Purpose

> **DEFINITION: SERVICE LEVEL MANAGEMENT (SLM)**
>
> The practice of setting clear business-based targets for service levels, and ensuring that delivery is assessed, monitored, and managed against these targets.

Key Agreements

- **Service Level Agreement (SLA):** Agreement between Provider and Customer. (e.g., "99.9% uptime").
- **Operational Level Agreement (OLA):** Agreement between internal IT teams. (e.g., Network team promises Server team 2ms latency).
- **Underpinning Contract (UC):** Agreement with an external supplier. (e.g., ISP promises internet access).

The "Watermelon SLA" Effect

A common failure in SLM is having SLAs that are **Green** (metrics met) on the outside, but **Red** (customer unhappy) on the inside.

Example: Server uptime is 99.99% (Green), but the server was down for the 1 critical hour during end-of-month processing (Red).

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Solution: Create **Business-Based Metrics**, not just technical ones. Measure user experience, not just server ping.

CASE STUDY: MANUFACTURING COMPANY MISALIGNMENT

Problem: IT reported 99.5% availability. Business was furious.

Why? The 0.5% downtime always happened during the shift change when all data was uploaded. The SLA didn't account for *critical business hours*.

Fix: Revised SLA to "100% availability required between 6am-8am and 6pm-8pm." Technical maintenance moved to midnight.

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Practice 6: Change Enablement (M4.L8)

Definition and Purpose

DEFINITION: CHANGE ENABLEMENT

The practice of maximizing the number of successful service and product changes by ensuring that risks have been properly assessed, authorizing changes to proceed, and managing the change schedule.

Note: It is "Change Enablement," not "Change Management." The focus is on enabling changes to happen fast and safely, not just blocking them.

Types of Changes

Change Type	Description	Authorization
Standard Change	Low risk, pre-authorized, well-understood, documented. (e.g., OS patch, laptop deployment).	Pre-authorized. No delay.
Normal Change	Needs to be scheduled, assessed, and authorized. (e.g., Major software upgrade).	Authorized by "Change Authority" (Manager, CAB, or Peer).
Emergency Change	Must be implemented ASAP (e.g., to fix a major incident or security breach).	Expedited authority (ECAB/Emergency Change Advisory Board).

CASE STUDY: 2022 BANK URGENT CHANGE DISASTER

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Scenario: A bank needed to update a regulatory report by a deadline.

Failure: They rushed it as an Emergency Change but skipped testing. The change contained a bug that blocked all credit card transactions for 6 hours.

Lesson: Emergency changes prioritize speed, but *some* testing/risk assessment is still required. Fast doesn't mean reckless.

Change Authority

The person or group who authorizes a change. In high-velocity IT (DevOps), the change authority is often the developer themselves (via automated testing), not a committee.

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Practice 7: Monitoring and Event Management (M4.L9)

Definition

The practice of systematically observing services and service components, and recording and reporting selected changes of state identified as events.

Event Types

- **Informational:** Just a log. "Backup completed successfully."
- **Warning:** Potential issue. "Server disk is 85% full." (Trigger proactive action).
- **Exception:** Something failed. "Server is down." (Triggers Incident Management).

Active vs. Passive Monitoring

- **Active:** Tools ping the device to check status.
- **Passive:** Tools wait for the device to send an alert.

> **CASE STUDY: MICROSOFT 2024 INCIDENT ANALYSIS**
>
> Microsoft found that **78%** of incidents had detectable warning signals (events) before the crash, but they were ignored.
>
> **Solution:** Better **Event Correlation** (AIOps). Using AI to spot patterns in warning events to prevent the exception event.

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Practice 8: IT Asset Management (M4.L10)

Definition

The practice of planning and managing the full lifecycle of all IT assets, to help the organization maximize value, control costs, manage risks, support decision-making about purchase, reuse, retirement, and disposal of assets.

Key Concept: IT Asset

Any financially valuable component that can contribute to the delivery of an IT product or service. (Servers, software licenses, cloud credits).

Lifecycle

Plan -> Acquire -> Deploy -> Maintain -> Retire -> Dispose.

CASE STUDY: SOFTWARE LICENSE FINE

A company didn't track software installs (Assets). An audit revealed they had 500 copies of Adobe Photoshop but only paid for 100.

Result: $2.8 Million fine. Effective ITAM would have prevented this by reconciling licenses vs. installs.

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Practice 9: Release Management (M4.L11)

Definition

The practice of making new and changed services and features available for use.

> **DISTINCTION: RELEASE VS. DEPLOYMENT**
>
> **Deployment:** Moving the code/hardware to the live environment. (Technical).
> **Release:** Making it available to users. (Business/Strategic).
> *You can deploy code (install it) but keep it hidden (Dark Launch) until you are ready to Release it (turn it on).*

Release Methods

- **Big Bang:** Release to everyone at once. (High risk).
- **Phased:** Release to Region A, then Region B.
- **Canary:** Release to 1% of users, check for bugs, then expand.

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Practice 10: Deployment Management (M4.L12)

Definition

The practice of moving new or changed hardware, software, documentation, processes, or any other component to live environments.

CASE STUDY: AIRLINE CHECK-IN SYSTEM

Deployment: Technicians installed the new software on servers in Tokyo, London, and NY.

Release: The airline announced the new feature to passengers.

Failure: The deployment was phased (hybrid mode), but the old and new systems couldn't talk to each other. Data inconsistency caused flight delays.

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Practice 11: Knowledge Management (M4.L13)

Definition

The practice of maintaining and improving the effective, efficient, and convenient use of information and knowledge across the organization.

Tacit vs. Explicit Knowledge

- **Tacit:** In your head. Experience. Hard to share.
- **Explicit:** Written down. Manuals. Easy to share.

Goal: Turn Tacit into Explicit.

> **CASE STUDY: THE RETIRED ENGINEER**
>
> A pharma company relied on one senior engineer to fix a specific machine. He retired. No one captured his knowledge.
>
> **Result:** Machine broke. Production stopped for 4 days. $3.7M loss. Knowledge Management ensures knowledge stays when people leave.

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Practice 12: Service Configuration Management (M4.L14)

Definition

The practice of ensuring that accurate and reliable information about the configuration of services, and the configuration items (CIs) that support them, is available when and where it is needed.

Configuration Item (CI) vs. Asset

- **Asset:** Has financial value (Cost, Depreciation).
- **CI:** Has operational value/dependencies (Relationships).

Example: A server is an Asset (worth $5k). It is also a CI (Hosts App A, Depends on Network B).

CMDB (Configuration Management Database)

The database that stores CIs and their relationships (Service Maps).

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Practice 13: Service Continuity Management (M4.L15)

Definition

The practice of ensuring that the availability and performance of a service are maintained at sufficient levels in case of a disaster.

RTO and RPO

- **Recovery Time Objective (RTO):** How fast must we be back up? (Time to recover).
- **Recovery Point Objective (RPO):** How much data can we lose? (Time since last backup).

CASE STUDY: HURRICANE HELENA INSURANCE DISASTER

An insurance company had a backup datacenter in the *same region* as their primary.

Result: Hurricane hit both. RTO was 4 hours; actual recovery took 11 days. Loss: $180M.

Lesson: Geographic diversity is key for continuity.

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Practice 14: Availability Management (M4.L16)

Definition

The practice of ensuring that services deliver agreed levels of availability to meet the needs of customers and users.

Metrics

- **MTBF (Mean Time Between Failures):** Reliability. (How long does it run before breaking?).
- **MTRS (Mean Time to Restore Service):** Maintainability. (How fast do we fix it?).

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Practice 15: Capacity and Performance Management (M4.L17)

Definition

The practice of ensuring that services achieve agreed and expected performance levels, satisfying current and future demand in a cost-effective way.

Sub-practices

- **Business Capacity Management:** Planning for future business growth (e.g., M&A).
- **Service Capacity Management:** End-to-end service performance.
- **Component Capacity Management:** CPU, RAM, Disk space tuning.

CASE STUDY: BLACK FRIDAY CRASH

Retailer added 40% more servers for Black Friday (Component Capacity). But the mobile app code was inefficient (Software Performance). The app crashed despite having enough servers. Capacity was fine; Performance was the bottleneck.

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Module Summary & Exam Prep

Exam Preparation Guide

HOW TO IDENTIFY PRACTICES IN QUESTIONS

- **Restoring service fast?** -> Incident Management
- **Finding root cause?** -> Problem Management
- **Routine user request (password/access)?** -> Service Request Management
- **Communicating with users/Single point of contact?** -> Service Desk
- **Assessing risk of an update?** -> Change Enablement
- **Moving code to production?** -> Deployment Management
- **Tracking relationships/dependencies?** -> Service Configuration Management
- **Tracking value/cost of hardware?** -> IT Asset Management
- **Disaster recovery?** -> Service Continuity Management
- **Negotiating uptime targets?** -> Service Level Management

Practice Questions Scenarios

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Q1: A user calls to ask how to use a feature in the new HR app. Which practice handles this?

A: Service Desk (Information Provision).

Q2: A server is running out of disk space. A warning alert is generated. Which practice?

A: Monitoring and Event Management.

Q3: The team investigates why the disk fills up every week to fix it permanently. Which practice?

A: Problem Management.

End of Module 4 Notes

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